

SECTION A

Q1] Select and write most appropriate answer from the given alternative

[16]

- 1) If the angles of a triangle are in the ratio 1:2:3, then the smallest angle in radian is
A) $\pi/3$ B) $\pi/6$ C) $\pi/2$ D) π
- 2) The cotangent of the angles $\pi/3$, $\pi/4$ and $\pi/6$ are in
A) A.P. B) G.P. C) H.P. D) Not in progression
- 3) The numerical value of $\tan 20^\circ \tan 80^\circ \cot 50^\circ$ is equal to.....
A) $\sqrt{3}$ B) $1/\sqrt{3}$ C) $2/\sqrt{3}$ D) $1/2\sqrt{3}$
- 4) If A(0,0), B(1,3) and C(k,0) are vertices of triangle ABC whose area is 3 sq.units then value of k is
A) 2 B) -3 C) 3 or -3 D) -2 or +2
- 5) Equation of a circle which passes through (3, 6) and touches the axes is
(A) $x^2 + y^2 + 6x + 6y + 3 = 0$ (B) $x^2 + y^2 - 6x - 6y - 9 = 0$
(C) $x^2 + y^2 - 6x - 6y + 9 = 0$ (D) $x^2 + y^2 - 6x + 6y - 3 = 0$
- 6) $i^2 + 1 + i + i^3 = \dots$

- A) 0 B) 1 C) -1 D) 1

7) If for a G.P. $t_6 / t_3 = 1458 / 54$ then $r = ?$

- A) 3 B) 2 C) 1 D) -1

8) In how many ways 4 boys and 3 girls can be seated in a row so that they are alternate.

- A) 12 B) 288 C) 144 D) 256

Q2] Answer the following questions

[04]

- i) Find the value of : w^{18}
- ii) If $A = \{1, 2\}$, find $A \times A$
- iii) Evaluate $f(x) = 2x^2 - 3x + 4$ at $x = 7$
- iv) Find the derivatives of the following functions of $y = \log x - \cos x$

SECTION B

Attempt any EIGHT of the following question

[16]

- Q3) Differentiate the following w.r.t. x. If $y = x^2 \sqrt{x} + x^4 \log x$
- Q4) If $f(x) = x^2 + 2$ and $g(x) = 5x - 8$, then find i) $(f+g)(1)$ ii) $(f-g)(-2)$
- Q5) If $t(m) = 3m^2 - m$ and $t(m) = 4$, then find m
- Q6) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set, $A = \{1, 3, 5, 7, 9\}$ $B = \{2, 3, 4, 6, 8, 10\}$,
 $C = \{6, 7, 8, 9\}$ find i) $(A')'$ ii) $(B-C)'$
- Q7) Find the value of $5P_2$
- Q8) Write in terms of factorials i) $5 \times 6 \times 7 \times 8 \times 9 \times 10$ ii) $3 \times 6 \times 9 \times 12 \times 15$
- Q9) For a G.P. if $a=6$, $r=2$, find S_{10} .
- Q10) Evaluate : $i^{30} + i^{40} + i^{50} + i^{60}$
- Q11) Convert following angles into degrees : $(\frac{-5\pi}{3})^c$
- Q12) The area of a circle is 225π sq. cm. Find the length of its arc subtending an angle of 120° at the centre. Also find the area of the corresponding sector.
- Q13) For $\theta = 30^\circ$, Verify that $\sin 2\theta = 2\sin\theta \cos\theta$

Q14) Find Minors and Cofactors of the elements of determinant $\begin{vmatrix} 2 & -3 \\ 4 & 7 \end{vmatrix}$

SECTION C

Attempt any EIGHT of the following question.

[24]

Q15) By using Cramer's rule solve the following linear equations. $x + y - z = 1$, $8x + 3y - 6z = 1$, $-4x - y + 3z = 1$

Q16) Find the equation of a circle whose centre is $(-3, 1)$ and which pass through the point $(5, 2)$.

Q17) One angle of a quadrilateral is $2\pi/9$ radian and the measures of the other three angles are in the ratio 3:5:8, find their measures in degree.

Q18) Show that $\frac{\sin(x+y)}{\sin(x-y)} = \frac{\tan x + \tan y}{\tan x - \tan y}$

Q19) If w is a complex cube root of unity, then show that

$$\text{i) } (1-w+w^2)^5 + (1+w-w^2)^5 = 32 \quad \text{ii) } (1-w)(1-w^2)(1-w^4)(1-w^5) = 9$$

Q20) Show that $\frac{12!}{5!7!} + \frac{12!}{6!6!} = \frac{13!}{6!7!}$

Q21) How many arrangements can be made, with the letters of the word CALCULATOR? In how many of these arrangements, vowels occur together?

Q22) Let X be the universal set, for the nonempty sets A and B , verify the De Morgan's laws

$$\text{i) } (A \cup B)' = A' \cap B' \quad \text{ii) } (A \cap B)' = A' \cup B'$$

Q23) Find the domain and range of $f(x) = \sqrt{9 - x^2}$

Q24) Derivative of Product of functions. If u and v are differentiable functions of x such that $y = u \cdot v$

$$\text{then, } \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Q25) Find the derivatives of the following functions $f(x) = x^5 \operatorname{cosec} x + \sqrt{x} \tan x$

Q26) Show that the function $f(x)$ is differentiable at $x = -3$ where, $f(x) = x^2 + 2$

SECTION D

Attempt any FIVE of the following question.

[20]

Q27) Find the value of (i) $7C_3$ (ii) $10C_7$ (iii) $52C_3$

Q28) i) One angle of a quadrilateral is $2\pi/9$ radian and the measures of the other three angles are in the ratio 3:5:8, find their measures in degree.

ii) Convert the degree 70° measures in the radian measures.

Q29) Prove that $\frac{\sin \theta}{1 - \cos \theta} + \frac{\tan \theta}{1 + \cos \theta} = \sec \theta \operatorname{cosec} \theta + \cot \theta$

Q30) In $\triangle ABC$ prove that i) $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$

Q31) If $A = \begin{bmatrix} 2 & 3 & -7 \\ 4 & 7 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 6 & -1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -1 & 6 \\ 0 & 2 & -5 \end{bmatrix}$, find the matrix X such that $3A - 2B + 4X = 5C$.

Q32) Find the equation of the circle whose centre is at $(3, -4)$ and the line $3x - 4y - 5 = 0$ cuts the circle at A and B ; where $l(AB) = 6$.

Q33) Find the derivatives of the following from the definition, (i) $\sqrt{x}$

Q34) If $z = 3 + 5i$ then draw Argand's diagram.